

Jev versus 7B Language Models for Speech-Neuroprosthesis Rescoring: A Pre-Registered Comparison in Two Participants

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Abstract

Speech neuroprostheses decode attempted speech from intracortical activity and end by rescoring the decoder’s candidate sentences with a language model of several billion parameters, the only component that needs a GPU. A cheaper rescorer must still pick the right sentence among near-identical candidates, and general language models asked to choose from a list in one call answer from where a label sits rather than from the sentence. We pose rescoring as a single typed decision served by Jev, a hosted model trained for calibrated decisions, and compare it with OPT-6.7b and Qwen2.5-7B on identical candidate lists from two participants with amyotrophic lateral sclerosis under two tuning protocols. On participant T15 (8.1% first-pass word error) Jev is ahead of both 7B models in all four comparisons and at most 0.20 points behind at the 95% bound. A pre-registered replication on participant T12 (19.2% first-pass word error) meets the fixed margin in two of four comparisons; the other two are inconclusive, with point estimates of +0.10 and +0.08 points. Open models given the same single-call task fail because they answer from label position. Jev costs \$0.07 to \$0.12 per thousand sentences and spends 62 ms at the provider per call, with end-to-end latency of 240 to 280 ms dominated by the network.

1 Introduction

A speech neuroprosthesis turns intracortical activity into text in three stages. A recurrent network trained with connectionist temporal classification (Graves et al., 2006) emits phoneme probabilities, a lexicon- and n -gram-constrained decoder turns them into a ranked list of candidate sentences, and a language model rescoring that list before the system speaks the winner (Willett et al., 2023a; Card et al., 2024a). This last stage is the only one that needs a multi-billion-parameter

model, so it sets the accelerator, the power budget and the cost of the system.

The rescorer’s job is selection: the answer is one of N sentences already written down. One call over the whole list is the natural form for a selector, but general language models handle it badly: asked to choose from a list, they favour positions over contents (Tang et al., 2023; Bito et al., 2026b).

A typed-decision model is trained for exactly this output. Given a state and a question with a fixed set of answers, it returns a probability for each answer in one pass and cannot produce anything outside the set. Jev (TypeSafe AI, 2026) is such a model, served through an API. Third-party evaluations report lower latency and cost than API-served language models on service orchestration (Li et al., 2026) and text annotation (Ibrahim and Zaki, 2026); none tests it on neural decoding or against scorers on a local GPU.

We replace the rescorer of the published pipeline with Jev for two participants and compare it with the two 7B models the stage would otherwise use, on byte-identical candidate lists. We derived the margin that defines *matches* from the first participant and fixed it, together with the full analysis, in a public pre-registration before decoding any data from the second. Figure 1 summarises the outcome.

Contributions.

- On T15, Jev is ahead of OPT-6.7b and Qwen2.5-7B in all four comparisons; on T12 the pre-registered test confirms two of four, and no point estimate favours a 7B model by more than 0.10 points (Section 4.1).
- Open models given the same single-call task fail on both participants: when they deviate from the decoder’s top candidate they pick the truth 0.2–2.4% of the time, against 15–

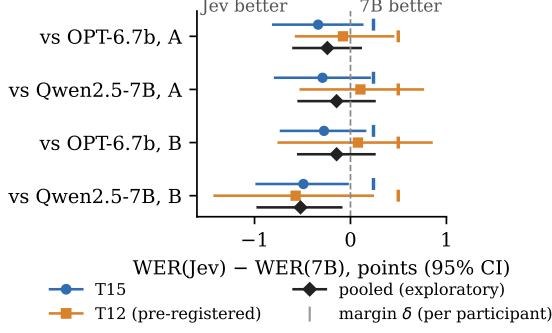


Figure 1: Jev minus each 7B rescorer on the test split, with paired 95% intervals. Ticks mark each participant’s margin δ . Points left of zero favour Jev.

27% for Jev (Section 4.2).

- Jev needs no local accelerator and costs \$0.07–0.12 per thousand sentences; its latency is network-bound and not below a local 7B scorer (Section 4.3).

We release candidate lists, per-arm scores, latency logs, the T12 decoder code and the pre-registration.¹

2 Related work

Willett et al. (2023a) and Card et al. (2024a) both finish decoding with a large language model, the latter in a system with single-digit error, as do the Brain-to-Text ’24 entries (Willett et al., 2024). Speech recognition interpolates the first-pass score with a language model (Udagawa et al., 2022) or trains the rescorer discriminatively (Xu et al., 2022; Shivakumar et al., 2023; Yu et al., 2023); discriminative training needs more lists than a few hundred development sentences supply, so every arm here, Jev included, runs without task-specific training.

Listwise reranking with language models suffers position bias, which permutation self-consistency (Tang et al., 2023) and order-invariant architectures (Bito et al., 2026a) address. Self-consistency multiplies the number of calls by the number of permutations, which removes the single-call property that motivates the typed format, so we do not apply it; we test open models in Jev’s single-call format and diagnose their failure. Prior Jev evaluations (Li et al., 2026; Ibrahim and Zaki, 2026) compare against generation over

¹<https://github.com/gabrycina/how-much-language-model>

an API; our baselines score on a local GPU, the harder comparison for latency.

3 Setup

Problem. For each attempted sentence the decoder emits a list H ranked by $\ell(h) = \lambda a(h) + g(h)$, where a is the acoustic score, g the n -gram score and λ the acoustic weight. A rescorer assigns $r(h)$ to every candidate and the system speaks

$$\hat{h} = \arg \max_{h \in H} (1 - \alpha) z_H[\ell](h) + \alpha z_H[r](h), \quad (1)$$

where z_H standardises scores across the list. A *per-candidate* rescorer sets $r(h)$ to the length-normalised log-likelihood of h ; a *typed* rescorer sees the list once, with h_i under label i , and we set $r(h_i) = \log \max(p_i, 10^{-6})$ for its label distribution p .

Data and lists. Both public datasets record a participant with amyotrophic lateral sclerosis attempting prompted sentences. For **T15** (Card et al., 2024a,b) we use the benchmark’s validation split (1,397 sentences) and the authors’ pre-trained phoneme network. For **T12** (Willett et al., 2023a,b) we use the test partition (877 unique sentences) and train the public baseline network ourselves, selecting the checkpoint on held-out train trials rather than on test as the baseline’s trainer does (Appendix B). For both, the Kaldi weighted finite-state transducer decoder (Povey et al., 2011) and OpenWebText 3-gram released with the T15 benchmark produce 100-best lists with identical options; the truth is in the list for 89% (T15) and 71% (T12) of sentences. Every arm scores the same cached lists.

Arms. We call each rescorer under test an *arm*. OPT-6.7b (Zhang et al., 2022) and Qwen2.5-7B (Qwen Team, 2024) score every candidate on one NVIDIA L40S. Jev (jev-latest, resolving to jev-1.13.0) receives each list once (Appendix A). As controls, Laya (Kishor, 2026), an open 421M typed-decision model, receives the identical request, and OPT-6.7b and Qwen2.5-7B receive the same task as a text prompt, read from their next-token probability of each label.

Protocols and statistics. We split each participant’s sentences 30/70 into development and test with a fixed seed (419/978 for T15, 263/614 for T12). Under protocol A we set λ to the value

Arm	T15 ($n=978$)		T12 ($n=614$)	
	A	B	A	B
No rescoring	8.09	8.09	19.23	19.23
OPT-6.7b	7.80	7.20*	18.84	17.59*
Qwen2.5-7B	7.75	7.42*	18.66	18.24*
Jev	7.46*	6.93*	18.76	17.67*

Table 1: Test word error (%). *: better than no rescoring after Holm correction over the three arms within a protocol and participant ($p < 0.05$, one-sided paired bootstrap). Full per-arm settings in Appendix C.

that minimises first-pass development error and tune α per arm; under **B** we tune λ and α jointly per arm. We score the test split once and report corpus-level word error rate (WER). Arm differences are paired percentile bootstraps over test sentences (5,000 resamples, two-sided 95% intervals). Jev *matches* a 7B model when the upper bound on $\text{WER}_{\text{Jev}} - \text{WER}_{7\text{B}}$ lies below $\delta = 0.025 \times$ the first-pass development error, which equals 0.24 points for T15 and 0.50 for T12. We chose δ after seeing T15, so T15 is exploratory; we committed the T12 analysis publicly before decoding any T12 data (Appendix E).

4 Results

4.1 Accuracy

On T15, Jev is the most accurate arm under both protocols (Table 1) and ahead of both 7B models in all four comparisons, by 0.28 to 0.49 points (Figure 1). Every interval excludes a Jev deficit larger than 0.20 points, inside the 0.24-point margin.

On T12, the pre-registered test confirms the margin against OPT-6.7b under protocol A (difference -0.08 , interval $[-0.57, +0.44]$) and against Qwen2.5-7B under protocol B (-0.57 , $[-1.42, +0.23]$). The other two comparisons are inconclusive: their point estimates are $+0.10$ and $+0.08$ points, but their intervals reach $+0.75$ and $+0.84$, above the 0.50-point margin. A plausible reading is that these failures reflect interval width on 614 noisier test sentences rather than a deficit, since neither point estimate exceeds 0.10 points. Under protocol B every main arm improves on no rescoring on both participants, by 0.67 to 1.64 points.

Pooling both participants, an analysis we did not pre-register, Jev is ahead in all four comparisons and at most 0.25 points behind at the 95% bound. Removing test sentences whose text also

Typed arm	T15			T12		
	used	on A	$\neq A$	used	on A	$\neq A$
Jev	57	72%	26.6%	69	57%	15.1%
Laya	44	69%	2.3%	51	49%	2.2%
Qwen2.5-7B	31	28%	2.4%	25	4%	1.1%
OPT-6.7b	13	35%	0.2%	11	20%	0.4%

Table 2: Typed arms before combination, all lists. *used*: distinct labels chosen out of 100; *on A*: share of answers on the decoder’s top candidate; $\neq A$: how often an answer other than A is the true sentence.

	T15	T12
Mean candidates per list	39	80
Jev input tokens per list, mean	1,631	2,825
Jev cost per 1,000 sentences	\$0.069	\$0.119
OPT-6.7b cost per 1,000 sentences, busy L40S	\$0.029	\$0.051
GPU break-even utilisation vs. Jev (OPT-6.7b)	43%	43%
Jev median latency, scoring run (ms)	282	253
OPT-6.7b median latency, scoring run (ms)	27	116
Jev end-to-end, London client (ms)	262	242
Jev server time, provider proxy (ms)	62	62
Jev two-option request end-to-end (ms)	253	221

Table 3: Cost and latency. Token counts from the API’s usage field on 140 (T15) and 60 (T12) random test lists; GPU cost at \$1.82 per L40S-hour. Scoring-run latencies are per list from a data centre in Finland; the London measurements pair each full request with a two-option request on the same keep-alive connection.

occurs in the phoneme network’s training data (107 for T15, 19 for T12) leaves every point estimate at or below $+0.03$ and widens the intervals (Appendix F).

4.2 The typed format needs a trained model

The gain comes from the model, not from the single-call format. Given the same task, Laya and the two 7B models in typed mode never beat no rescoring on either participant (Appendix C). All four typed arms show strong position preferences, but not for the same position. Jev and Laya place 49–72% of their answers on label A, the decoder’s own top candidate. The open 7B models instead concentrate on fixed positions whatever the list contains: OPT-6.7b gives 27% of its T15 answers and 61% of its T12 answers to position 68, and Qwen2.5-7B gives 16% and 26% to position 27. What separates the arms is what happens when they deviate from label A (Table 2): Jev’s other answers are the true sentence 15–27% of the time, the open models’ 0.2–2.4%.

4.3 Cost and latency

Jev bills a mean 1,631 input tokens per T15 list and 2,825 per T12 list, which is \$0.069 and \$0.119

per thousand sentences. OPT-6.7b on a fully used L40S costs \$0.029 and \$0.051, so a busy GPU is cheaper per sentence; the GPU wins only above 43% utilisation on both participants. One user produces a small fraction of a GPU’s hourly capacity of about 36,000 to 62,000 lists: at 1,000 sentences a day Jev costs \$0.07 to \$0.12, against \$43.68 for a dedicated on-demand GPU.

Jev answers in a median 282 ms (T15) and 253 ms (T12), against 27 ms and 116 ms for OPT-6.7b on the local GPU. The provider’s proxy reports a median 62 ms of server time on both participants, independent of list length, and a two-option request takes almost as long end to end as a full list (Table 3). The remaining 175 to 195 ms is transport. A co-located deployment would approach the same order as a local 7B model, not below it; we could not test this.

5 Conclusion

One typed decision from Jev rescues a speech neuroprosthesis’s candidate list about as well as a 7B language model that scores every candidate: ahead in all four comparisons on T15 and in the pooled data, and within a pre-registered margin in two of four on T12, with no point estimate favouring the 7B models by more than 0.10 points. Open models given the same task fail, so the result depends on a model trained for the decision. The next step is an open typed-decision model of known size trained on this task, which would let the cost and latency claims be tested on a patient’s own hardware and allow a confirmatory test with more participants.

Limitations

One closed model. Jev’s size and architecture are undisclosed, so the result shows that a typed-decision model does this job, not why; hosted models also change over time. We release its per-sentence label probabilities so the analysis is reproducible, though the model is not. **Two participants, limited power.** Differences below about half a point are hard to resolve, which is why two T12 comparisons are inconclusive. **Margin.** We derived δ from T15; only T12 is confirmatory. **Our T12 system is a baseline.** We trained the public baseline network and used the T15 benchmark’s 3-gram decoder, not the published T12 system, so absolute T12 error rates are not comparable to the original study. **Cloud dependence.** A

hosted rescorer sends decoded speech to a third party and fails without a network; a clinical device would need a private deployment, whose latency we could not measure. **Prompt wording.** The typed prompt asks for the most fluent sentence; we tested one wording.

Ethics statement

Both datasets were collected under the original studies’ clinical-trial approvals and consent procedures (Willett et al., 2023a; Card et al., 2024a) and are released publicly without identifying information; we collected no new data. A hosted rescorer sends a user’s decoded speech to a third party, which raises privacy and availability concerns for a person who depends on the device to communicate; our cost and latency results should not be read as a recommendation to deploy a cloud service without addressing them. The author has no financial or professional relationship with Type-Safe AI. The typed format limits one risk: the rescorer only chooses among sentences the decoder proposed, so it cannot insert words the user did not attempt.

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A Prompts

Jev and Laya receive one request per list through the same interface, with candidates in the decoder’s order under labels A, B, and so on.

```
state = {"task": "A speech decoder produced these candidate
sentences for what a person was trying to say.
Exactly one is what they meant.",
"candidates": {"A": "<cand. 1>", "B": "<cand. 2>", ...}}
questions = {"sentence": {"type": "choice",
"instructions": "Which candidate is the sentence the person
actually meant? Judge by which reads as fluent, natural
English.",
"criteria": {"A": "<cand. 1>", "B": "<cand. 2>", ...}}}
```

OPT-6.7b and Qwen2.5-7B receive the following text, wrapped in the model’s chat template where one exists and followed by The answer is; we read each label’s next-token probability.

A speech decoder produced these candidate sentences for what a person was trying to say. Exactly one is what they meant. Answer with the single label of the most fluent, natural English sentence.

```
A: <candidate 1>
B: <candidate 2>
...
Answer:
```

B T12 phoneme network

We follow the public baseline for this dataset (Willett et al., 2024): threshold crossings and spike-band power from the 128 area-6v electrodes (256 features), z-scored per recording block; a day-specific input layer followed by a five-layer bidirectional GRU with 1,024 units, a kernel of 32 bins and a stride of 4; CTC over 39 phonemes, a word-boundary token and a blank; 10,000 batches of 64 with Adam (learning rate 0.02, $\epsilon = 0.1$, weight decay 10^{-5}), dropout 0.4, white-noise (s.d. 0.8) and constant-offset (s.d. 0.2) augmentation, seed 0. Phoneme targets come from g2p_en. The baseline’s trainer keeps the checkpoint with the lowest phoneme error on the test partition, which would leak the evaluation set into model selection. We hold out 10% of each session’s train trials (880 trials; 7,920 remain for training) and select on them. The best checkpoint reaches 20.4%

phoneme error on this validation set; the test partition is read only to produce the evaluation lists.

C Full results

Arm	Pr.	λ	α	WER	Δ	p	p_{Holm}
No rescoring		0.3	—	8.09	—	—	—
OPT-6.7b	A	0.3	0.30	7.80	-0.29	0.035	0.071
OPT-6.7b	B	0.4	0.45	7.20	-0.89	<0.001	<0.001
Qwen2.5-7B	A	0.3	0.45	7.75	-0.34	0.064	0.071
Qwen2.5-7B	B	0.45	0.45	7.42	-0.67	0.002	0.002
Jev	A	0.3	0.90	7.46	-0.63	0.006	0.017
Jev	B	2	0.75	6.93	-1.16	<0.001	<0.001
OPT-6.7b (typed)	A	0.3	0.00	8.09	+0.00	1.000	—
OPT-6.7b (typed)	B	0.25	0.20	8.37	+0.28	1.000	—
Qwen2.5-7B (typed)	A	0.3	0.20	8.11	+0.02	0.576	—
Qwen2.5-7B (typed)	B	0.3	0.20	8.11	+0.02	0.576	—
Laya (typed)	A	0.3	0.00	8.09	+0.00	1.000	—
Laya (typed)	B	0.3	0.00	8.09	+0.00	1.000	—

Table 4: T15: every arm under both protocols, test split ($n=978$). Δ and p against no rescoring (one-sided paired bootstrap); Holm over the three main arms within a protocol.

Arm	Pr.	λ	α	WER	Δ	p	p_{Holm}
No rescoring		0.45	—	19.23	—	—	—
OPT-6.7b	A	0.45	0.20	18.84	-0.39	0.080	0.116
OPT-6.7b	B	1	0.25	17.59	-1.64	<0.001	<0.001
Qwen2.5-7B	A	0.45	0.30	18.66	-0.57	0.058	0.116
Qwen2.5-7B	B	0.6	0.35	18.24	-0.99	0.007	0.007
Jev	A	0.45	0.15	18.76	-0.47	0.021	0.063
Jev	B	1.5	0.30	17.67	-1.56	<0.001	<0.001
OPT-6.7b (typed)	A	0.45	0.00	19.23	+0.00	1.000	—
OPT-6.7b (typed)	B	0.45	0.00	19.23	+0.00	1.000	—
Qwen2.5-7B (typed)	A	0.45	0.10	19.28	+0.05	0.664	—
Qwen2.5-7B (typed)	B	0.4	0.15	19.65	+0.42	0.965	—
Laya (typed)	A	0.45	0.00	19.23	+0.00	1.000	—
Laya (typed)	B	0.45	0.00	19.23	+0.00	1.000	—

Table 5: T12: every arm under both protocols, test split ($n=614$). Δ and p against no rescoring (one-sided paired bootstrap); Holm over the three main arms within a protocol.

D Cost and latency

The London measurement pairs each full request with a two-option request on the same keep-alive connection and records the provider’s x-envoy-upstream-service-time header; scoring-run latencies come from a data centre in Finland. Table 3 gives the figures.

E Pre-registration

Before any T12 data was decoded we committed to a public repository a document fixing the data partitions, the phoneme-network recipe and its selection rule, the decoder options, the 30/70 split and seed, the arms, both tuning protocols and their grids, the bootstrap procedure, the margin $\delta = 0.025 \times$ first-pass development error, and

Jev minus	all test	without overlap
T15 OPT-6.7b, A	-0.34 [-0.80, +0.12]	-0.28 [-0.78, +0.22]
T15 Qwen2.5-7B, A	-0.29 [-0.78, +0.20]	-0.24 [-0.77, +0.26]
T15 OPT-6.7b, B	-0.28 [-0.72, +0.15]	-0.19 [-0.65, +0.28]
T15 Qwen2.5-7B, B	-0.49 [-0.98, -0.03]	-0.45 [-0.96, +0.08]
T12 OPT-6.7b, A	-0.08 [-0.57, +0.44]	-0.11 [-0.60, +0.44]
T12 Qwen2.5-7B, A	+0.10 [-0.52, +0.75]	+0.03 [-0.59, +0.68]
T12 OPT-6.7b, B	+0.08 [-0.75, +0.84]	+0.00 [-0.81, +0.81]
T12 Qwen2.5-7B, B	-0.57 [-1.42, +0.23]	-0.72 [-1.54, +0.13]

Table 6: Primary comparisons with and without test sentences whose text occurs in the phoneme network’s training data (T15: 107 removed, $n=871$; T12: 19 removed, $n=595$).

the decision rule (confirmed if all four comparisons meet δ , partly confirmed if some do). We then committed the analysis code, checked against T15, before scoring any T12 list. The public repository records the order: the plan (commit 0c2a4a2, 29 September 2026, 20:45 UTC+1) precedes the download of the T12 data; the analysis code (5b88f4e, 21:08) and label diagnostic (b0958d1, 21:10) precede the T12 scores, which we committed with the results (de2d2a3, 21:51).

Deviations: (i) the plan said the Jev model version would be recorded at run time; our client did not store it per request, so we queried the API on the day of the run, when jev-latest resolved to jev-1.13.0. (ii) An exploratory open 0.6B typed-decision model was not run: it accepts at most 26 options and requires a custom package. (iii) The pooled two-participant analysis in Section 4.1 was added afterwards and is exploratory.

F Test sentences seen in training

Some test sentences also occur, as text, in the phoneme network’s training data (different recordings of the same sentence): 107 of 978 for T15 and 19 of 614 for T12. They are a property of the benchmarks’ splits. Table 6 repeats the primary comparisons without them: every point estimate stays at or below +0.03 points and the intervals widen.